Canto-L · TrackML

Track reconstruction at pileup 200 · frozen pipeline, evaluated once on 300 test events · each row under the reference's own metric code (ported, checked against the original)

	Canto-L 1 seed · TEST	EggNet Calafiura 2024 · Chan 2025	EC Calafiura et al. 2024	BC-HGNN Liu et al. 2026	EC-GNN Liu et al. 2026
Track efficiency TrackML-1GeV · EggNet code	**99.56%**	**99.56%**	98.98%	—	—
Fake-track rate TrackML-1GeV · EggNet code lower is better	**0.008%**	0.06%	0.12%	—	—
Duplicate rate TrackML-1GeV · EggNet code lower is better	**0.13%**	1.29%	4.21%	—	—
Track efficiency TrackML-1GeV · Liu et al. code	**99.77%**	—	—	98.55%	—
Fake rate TrackML-1GeV · Liu et al. code lower is better	**0.26%**	—	—	1.23%	—
Track efficiency full TrackML · EggNet code	**99.41%**	95.87%	—	—	—
Fake-track rate full TrackML · EggNet code lower is better	2.40%	**0.19%**	—	—	—
Duplicate rate full TrackML · EggNet code lower is better	**0.17%**	2.76%	—	—	—
Track efficiency full TrackML · Liu et al. code	**99.65%**	—	—	97.86%	96.35%
Fake rate full TrackML · Liu et al. code lower is better	**34.94%**	—	—	36.71%	55.58%

All numbers: TrackML throughput phase (Zenodo 4730157), pileup 200; only references that used the same phase are shown (established from the authors' code and released data). Canto-L: the frozen pipeline at working point WP-A, chosen on 150 val events (the highest DM efficiency of targets at a track fake rate ≤ 2.94 %, the HNL value, not re-tuned; configurations within one standard error tied, the lowest fake rate wins), evaluated once on 300 test events; one training seed. Every row is recomputed after the freeze from the frozen pipeline's test tracks with ports of the reference's evaluation code, checked against the unmodified code on hand-counted, randomised and real events (EggNet: github.com/exatrkx/EggNet e2bd117; Liu et al.: eval_metrics of github.com/ryanliu30/Tracking-ML-Exa.TrkX f107781, strict > 50 % majority, which reproduces their Truth-CC; HierarchicalGNN a2563af has >= 50 %); nothing was re-chosen. Event-bootstrap errors on the Canto-L values are at most 0.11 points. Published values as quoted, on the authors' own events; where no uncertainty is quoted (Liu et al., the Calafiura et al. fake rates) the 2-sigma test uses the Canto-L error alone. Bold and shaded: the best value of a row and every value within 2 sigma of it (not resolved).

Working points chosen on val for one reference each, with our readings of its definitions. Full TrackML, EggNet code, at the point chosen for Chan et al.'s fake rate: efficiency 99.37 %, duplicates 1.70 %, fake tracks 0.173 ± 0.003 % (EggNet 95.87 %, 2.76 %, 0.19 ± 0.01 %). TrackML-1GeV, at the point chosen for the fake rate of Calafiura et al.'s EC: efficiency 99.67 ± 0.01 %, duplicates 0.77 %, fake tracks 0.039 %. Full TrackML, Liu et al. code, at the point chosen for the FR of their EC-GNN: efficiency 99.66 ± 0.01 %, FR 35.46 ± 0.11 % (EC-GNN 96.35 %, 55.58 %).

Caveats. Liu et al. state that their FR is a model-to-model comparison, not final track-building performance; Canto-L's is after its final builder. Liu et al.'s code takes a particle's pT as the minimum hit-level truth pT, our tables hold the initial pT. Our TrackML-1GeV events keep the hits of particles with pT > 1 GeV (EggNet's hard cut; Liu et al.'s released events cut hits by hit-level pT). Ju et al. (Exa.TrkX) evaluated on the accuracy phase and is not compared.

Cross-phase test (300 accuracy-phase events, Kaggle train_1, whose digitisation differs: pixel clusters about twice as large): the same frozen Canto-L reaches 95.98 % target efficiency (test 99.77 %) and 0.80 % fake tracks (0.055 %) in TrackML-1GeV; without noise hits 94.67 % (99.66 %), 7.94 % (1.29 %) and a TrackML score of 0.614 (0.968), whose accuracy-phase weights also count the particles produced in the simulation (9.1 % of the weight; none in the throughput phase). All-hit results pending. Context, not a comparison: TrackML score on full test events 0.966; challenge winners: accuracy phase (Kaggle 2018) Top Quarks 0.92182, throughput phase (2 CPU cores) Mikado 94.4 % at 0.56 s per event.